



Qualcomm hit with fine

The European Union has fined Qualcomm by \$12 billion for rewarding Apple on exclusively using their chips. <http://digit.in/QualApp>



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brought out by ASUS, MSI, GIGABYTE, Inno3D, ZOTAC or EVGA since the performance would be the same. However, that's not the case. These manufacturers have shipped their cards with a monitoring/overclocking software. The high-performance profiles which were earlier baked into these cards can now be accessed via these software. And it's not necessary that you have to stick to a particular software since they all use the same NVIDIA API. We find MSI Afterburner to be the best, but we went ahead and installed the manufacturer recommended software to get the tried and tested performance profiles.

The other manner in which third-party cards offer more value is by giving you an improved cooling solution and a better designed PCB. These two factors, combined together, allow for higher stability while overclocking graphics cards. This is where we will be focussing our attention.

THE COMPETITORS

In one corner we have the ROG STRIX GTX 1070 Ti from ASUS. Sporting a triple-fan cooler, this GTX 1070 Ti is a tad thicker than your stock GTX 1070 Ti. However, you end up with shielding on the back, a custom PCB and a heatsink as big as the graphics card itself. And in the other corner we have the GTX 1070 Ti AMP EXTREME from ZOTAC. The AMP EXTREME cards have had a track record of having some of the highest factory-overclocks in the graphics card business, however, this being a 1070 Ti, that advantage is taken away. Both graphics cards start off with the same base and boost clocks of 1607 MHz and 1683 MHz, respectively. The ZOTAC card too sports a triple fan cooler and is a bit thicker than the ASUS card. Also, it has a custom PCB and a back cover. Let's take a closer look at the two.

POWER AND TEMPERATURE

ASUS ROG STRIX GTX 1070 TI

The ROG STRIX card uses only one 8-PIN ATX power connector which is more than sufficient for the 180 W power draw that the stock GTX 1070 Ti is designed for. This being a custom card will draw more since it also ships with two OC profiles within ASUS GPU Tweak. Even with the higher draw, the single 8-PIN connector is sufficient. As for the PCB, the ASUS ROG STRIX 1070 Ti makes use of seven power phases, six of which are used to power the GPU. It has a uP9511P voltage regulator set in a 6+1 phase configuration. Each choke is moulded within concrete which works great towards minimising coil whine.

The heatsink is tin-plated copper all over, including the contact surface. There are six heatpipes drawing the heat away from the GPU and the MOSFETs. The fans are all mounted atop a metal bracket that sits on the heatsink. The card idles at around 34–36 degrees Celsius and ramps up to 82 degrees at peak load. The ASUS card also has additional fan connectors coming out from the side allowing your GPU to regulate the load lines for system fans in tandem with itself.



ZOTAC GTX 1070 TI AMP EXTREME

The ZOTAC card has two 8-PIN connectors for drawing power, which is more than needed. However, one has to acknowledge the fact that having the power spread across multiple lines helps with lowering the heat generated in the wires and at the connector. The ZOTAC FireStorm software does not have multiple OC profiles for the GTX 1070 Ti like the ASUS card has. However, you can find custom overclocking profiles on ZOTAC's website which can be easily applied. It's not a user-friendly way, but at least there is a way. As for the PCB, this card too has a custom VRM circuitry. Like the ASUS card, ZOTAC also uses a uP9511 for the voltage regulator to control 8 phases. The memory phases are handled using a uP1666. The configurations of both cards are very similar but the ZOTAC card implements phase doubling.

The heatsink for the ZOTAC card is quite beefy as well and the contact surface is plain copper. Moreover, the VRM heatsink is separate from the main assembly. The card idles around 35 degrees Celsius with load temperatures hitting 81 degrees which is only a bit better off than the ASUS card.

